SWAPNIL SHARMA

Ph.D. Candidate in Economics · Boston College

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EDUCATION

Boston College, Boston, MA — *Expected May 2027*
Ph.D. in Economics (Econometrics and Quantitative Economics)

Indira Gandhi Institute of Development Research, Mumbai, India — *July 2019*
M.Phil. in Economics

Indira Gandhi Institute of Development Research, Mumbai, India — *May 2018*
M.Sc. in Economics

Indian Institute of Technology (BHU), Varanasi, India — *May 2012*
B.Tech. in Mechanical Engineering

RESEARCH FIELDS

Macroeconomics; Monetary Economics; Applied Econometrics; Theoretical Industrial Organization

PH.D. WORKING PAPERS

Credit Markets, Uncertainty, and Business Cycles *Job Market Paper* [Link to Paper]

Shows that the shock explaining most business-cycle variation in hours combines two forces with distinct propagation: macro uncertainty causes a broad contraction, while a credit-related shock produces a boom followed by a bust and accounts for the spectral peak in hours. An estimated financial-frictions model, solved with a Deep Equilibrium Network, reproduces both patterns through uncertainty about enforcement and noisy signals about future productivity.

Monetary Policy Surprise and Forward Information

Shows that roughly 10% of high-frequency policy shocks are predictable from forward information held by the Federal Reserve and forecasters. An orthogonalized shock series produces larger and more statistically significant macroeconomic effects than earlier high-frequency studies.

Forward-Looking Credit Pricing and the Endogenous Cycle in Hours Worked *with Sanjay Moorjani*

Shows that the endogenous nine-year cycle in hours and credit spreads disappears when lenders price household credit using expected default risk over the life of the loan rather than current employment. Re-estimation preserves the cycle only for loans lasting about 1-1.5 quarters; with longer loans, the best-fitting model relies on persistent external shocks.

PRE-PH.D. PAPERS

Competition in Two-Sided Markets with Congestion [Link to Paper] [IGIDR Working Paper]

Extends a two-sided platform model to include within-group congestion. Congestion raises platform profits when tolerance is low under single-homing, while the condition reverses under multi-homing.

First Party Content With Sellers Competition in Two-Sided Market [Link to Paper]

Studies how a monopoly platform chooses prices and first-party content when sellers compete. Seller competition can restrict seller participation and reverse standard predictions about first-party-content investment under unfavorable expectations.

Competition among Sellers and Platform Entry into Sellers' Product [Link to Paper]

Compares pure intermediation with a platform that competes in sellers' product market. Entry lowers sellers' membership fees and can benefit all sides under monopoly; with competing platforms, buyer gains depend on product differentiation and perceived quality.

RESEARCH EXPERIENCE

Research Statistician (Consultant), Research Services, Boston College — *Jun 2024–Aug 2026*

- Automated extraction, translation, and structuring of data from 1,000+ scanned Hindi PDFs into analysis-ready datasets.
- Applied machine learning to extract structured, privacy-safe transaction data from 3,000+ heterogeneous credit card statements.
- Consulted on 10+ academic projects involving data acquisition, cleaning, statistical analysis, and modeling.

Research Assistant to Professor Theodore Papageorgiou, Boston College — *Summer 2022*

- Developed and optimized Python algorithms for a high-dimensional fixed-point problem in a macro-labor equilibrium model.

TEACHING EXPERIENCE

Teaching Fellow (Lecturer), Boston College
- Macroeconomic Theory — *Fall 2024*
- Macroeconomic Theory — *Summer I 2025; Summer II 2025*

Teaching Assistant, Boston College
- Principles of Macroeconomics — *Fall 2019*
- International Finance — *Spring 2020; Fall 2023*
- Principles of Economics — *Fall 2022*
- Statistics — *Spring 2022, 2023, and 2024*
- Econometric Methods — *Fall 2026*

Teaching Assistant, IGIDR
- Microeconomic Theory — *Fall 2018*
- Game Theory — *Spring 2019*

ADDITIONAL PROFESSIONAL EXPERIENCE

Intern, Foreign Exchange Department, Reserve Bank of India — *Summer 2017*
- Supported assessment of a $150 million external commercial borrowing for compliance with foreign-exchange regulations.

Assistant Manager, Research and Development, Hero MotoCorp Ltd. — *2012–2014*
- Coordinated development and validation of new motorcycle components for four models.
- Inspected and approved 15+ vendor samples and supported production-capacity expansion.

HONORS AND SCHOLARSHIPS

Boston College, Tuition remission and stipend — *2019–Present*
Indian Institute of Technology Joint Entrance Examination, top 1% — *2008*

TECHNICAL SKILLS

Programming and Data: Python, MATLAB, SQL, Stata, Julia, Microsoft Excel
Agentic AI-assisted research workflows: pipeline development and automation with coding agents, integrated with Python, MATLAB, Stata, and LaTeX
Modeling and computation: DSGE Modeling, Global Solution Methods, Numerical Optimization, Neural Networks, Deep Learning

CERTIFICATIONS

Pricing Options with Mathematical Models, Caltech (Coursera) · *July 2026*
Pandas, Kaggle · *June 2026*

REFERENCES

Rosen Valchev (Chair)
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Peter Ireland
Murray and Monti Professor
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Susanto Basu
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